

When Temporal Differences Mislead: Reliability-Guided Evidence Alignment for Zero-Shot Remote-Sensing Change Reasoning

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Abstract. Change monitoring from satellite and aerial imagery supports climate adaptation, conservation planning, and post-disaster assessment, yet seasonal appearance, sensor differences, registration error, and off-nadir geometry produce discrepancies that resemble real change. We present **RGEA** (*Reliability-Guided Evidence Alignment*), a frozen-CLIP framework for zero-shot bi-temporal change analysis. RGEA matches visual features to changed/unchanged text prototypes, keeps the *direction* of temporal evidence with a signed residual, and learns from the discrepancy *magnitude* a reliability score that controls how strongly the evidence is used; swapping the observations therefore reverses evidence polarity while leaving reliability unchanged. Trained on LEVIR-CD only, RGEA reaches 89.04 pixel AUC and 83.48 PRO on unseen WHU-CD, while DSIFN-CD and S2Looking expose semantic-calibration and geometric-grounding failures, showing why Earth-observation systems should report both decision quality and spatial evidence under domain shift. Code: <https://github.com/leolin65/RVLCD>.

Keywords: Zero-shot change detection · domain generalization · vision-language models

1 Introduction

Satellite and aerial image streams support urban-expansion monitoring, post-disaster assessment, and other land-surface analyses, but vary across geography, season, viewpoint, resolution, and sensor. Dense labels are costly for every new region, so a model trained in one place is deployed in another, where it may respond to shadows, vegetation cycles, parallax, or registration error instead of structural change. Concretely, a half-pixel misalignment shifts every edge, a change of sun angle re-colours every roof, and a new building alters only a few hundred pixels; all three yield a large $T_2 - T_1$ difference, and an absolute-difference detector fires on all three. A pair-level alert is therefore not enough:

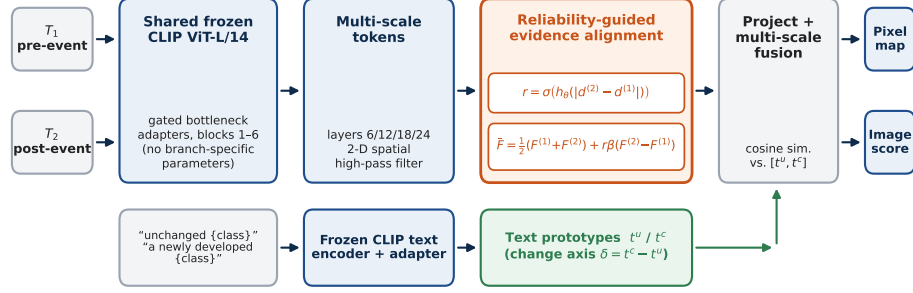


Fig. 1: RGEA overview. Both observations share one frozen CLIP encoder with gated adapters; the average keeps context, the signed residual carries evidence direction, and a magnitude-based reliability gate r limits uncertain differences before cosine matching to the changed/unchanged prototypes.

the dense response should coincide with the changed structure, unreliable discrepancies should be down-weighted, and the detector should know whether the second observation moved *toward* or *away from* change, which absolute differencing discards.

We propose **RGEA**, a reliability-guided evidence-alignment framework on frozen CLIP [3]. Text prototypes for *changed* and *unchanged* define a semantic axis, and the two observations are decomposed into an average context feature, a signed residual $F^{(2)} - F^{(1)}$ that carries the *direction* of temporal evidence, and a scalar reliability r computed only from the discrepancy *magnitude*, which sets how much of that evidence enters the prediction. The separation matters because a large difference may reflect a real transition or merely poor acquisition geometry.

Contributions: (i) a reliability-aware formulation of zero-shot change analysis separating pair decisions, dense localization, coverage, and threshold transfer; (ii) a signed-evidence pathway with a magnitude-only reliability gate and a temporal-order counterfactual; (iii) a source-only study with ablations on WHU-CD, DSIFN-CD, and S2Looking diagnosing semantic, calibration, and geometric failures.

2 Reliability-Guided Evidence Alignment

Semantic prototypes. Two prompt families, “unchanged {class}” and “a newly developed {class}” ($\{\text{class}\} = \textit{building}$ here), are encoded by the frozen CLIP text encoder and refined by a lightweight text adapter. The ℓ_2 -normalized prototypes t^c and t^u define the change axis $\delta = t^c - t^u$. The adapter is trained first (Stage 1) against frozen patch-level temporal magnitudes and the change mask, with an orthogonality term keeping t^c and t^u apart, then frozen while the image side is trained (Stage 2); two Stage-1 epochs suffice (Sect. 3).

Shared visual encoding. Both observations share one frozen CLIP ViT-L/14-336. Blocks 1–6 each receive a bottleneck adapter (down-projection, GELU, up-projection) blended with the block output through a learned sigmoid gate;

adapters and gates are shared by both temporal branches, so no branch-specific offset can masquerade as change. Patch tokens from layers 6, 12, 18, and 24 pass through a 2D spatial high-pass filter that suppresses scene-wide illumination and seasonal colour variation while keeping local structure such as roof edges.

Signed evidence and reliability. Let $d^{(1)}, d^{(2)}$ be the pooled global descriptors of the two observations. A two-layer MLP h_θ sees only the magnitude of their difference:

$$r = \sigma\left(h_\theta\left(|d^{(2)} - d^{(1)}|\right)\right) \in [0, 1], \quad (1)$$

and estimates how much the pair should be trusted. The semantic pathway keeps the signed residual at each scale ℓ ,

$$\bar{F}_\ell = \frac{F_\ell^{(1)} + F_\ell^{(2)}}{2} + r\beta\left(F_\ell^{(2)} - F_\ell^{(1)}\right), \quad (2)$$

with one learned scale β . When $r \rightarrow 0$ the fused feature collapses to the average context; when r is high it moves along the change direction, with β bounding how far. Multi-scale features are projected to the text space and combined with softmax-normalized (non-negative, sign-preserving) weights; cosine similarity to $[t^u, t^c]$ gives a two-channel dense map and an image score. Stage 2 trains adapters, gates, scale weights, h_θ , and β with focal/Dice losses on the dense map and cross-entropy on the image score.

Temporal-order audit. Swapping the two observations, $(T_1, T_2) \mapsto (T_2, T_1)$, preserves the average μ and the discrepancy magnitude, hence r , but reverses the residual, $\Delta \mapsto -\Delta$, so the evidence term $r\beta\langle\Delta, \delta\rangle$ changes sign while reliability is unchanged, whereas absolute differencing is invariant to the swap and cannot preserve polarity.

3 Cross-Domain Earth-Observation Evaluation

Datasets. LEVIR-CD [1] (637 Google-Earth pairs, 0.5 m, building change) is the only training domain; no target images, labels, fine-tuning, or model selection are used. WHU-CD [2] (aerial, Christchurch, 2012 vs. 2016, 0.075 m) keeps building-change semantics but changes sensor and resolution: *compatible-semantics* transfer. DSIFN-CD [5] (Google Earth, six Chinese cities) labels roads, buildings, cropland, and water, so a building prior covers only part of its labels: *semantic-interface* transfer. S2Looking [4] (side-looking satellite pairs, 0.5–0.8 m, rural) adds off-nadir parallax and occlusion: *geometric* transfer.

Metrics. *Pair ranking*: image-level AUC/AP (I-AUC, I-AP), positive if any pixel changed. *Dense ranking*: threshold-free pixel AUC/AP (P-AUC, P-AP). *Regional coverage*: per-region overlap (PRO), mean overlap with each connected changed region integrated over false-positive rate, so small structures count equally. *Threshold transfer*: pixel F1/IoU at one threshold chosen on the LEVIR-CD validation split and applied unchanged to every target, i.e. calibration rather than ranking.

Table 1: Zero-shot transfer, trained on LEVIR-CD only: pair ranking, dense ranking, regional coverage, and fixed-threshold calibration.

Target	I-AUC	I-AP	P-AUC	P-AP	PRO	P-F1	P-IoU
WHU-CD	84.02	75.25	89.04	61.51	83.48	60.00	42.85
DSIFN-CD	73.94	83.73	74.51	54.74	39.32	11.33	6.00
S2Looking	90.88	99.98	58.57	2.65	26.63	3.63	1.85

Results. Table 1 reads differently on each target. On WHU-CD all four questions are answered well (89.04 P-AUC, 61.51 P-AP, 83.48 PRO, 60.00 P-F1 at the transferred threshold): with compatible semantics the dense evidence is both well ranked and well calibrated. On DSIFN-CD dense ranking stays useful (74.51 P-AUC, 54.74 P-AP) while PRO falls to 39.32 and P-F1 to 11.33: changed pixels are still ordered correctly but scores are shifted and non-building regions only partly covered, a semantic-interface and calibration limit rather than a failure to see change. S2Looking shows the clearest reliability gap: pair ranking is high (90.88 I-AUC, 99.98 I-AP, inflated because almost every pair contains some change) while P-AP is 2.65 and PRO 26.63. Off-nadir parallax creates localized residuals that look like change, and one global reliability scalar can lower the pair’s trust but cannot suppress them selectively; S2Looking is thus a stress case motivating spatial, registration-aware reliability.

Ablations. Varying the Stage-1 text-adaptor schedule gives WHU-CD P-AUC/PRO of 88.12/78.84 with no adaptation, 85.74/80.94 after one epoch, **89.04/83.48** after two, 87.71/83.02 after three, and 86.38/81.01 after five: without adaptation the prototypes rank pixels well but cover regions poorly, two epochs are best on both, and longer schedules fit LEVIR-CD specifically, so text-side capacity trades in-domain fit against transferability within a narrow window. Replacing the constrained fusion of Eq. (2) with an MLP over $[F^{(1)}, F^{(2)}]$ or with a learned gate that reweights the branches freely did not improve dense localization on WHU-CD, and the gated variant was worst on the dense metrics, consistent with residual polarity no longer being guaranteed; the constrained form preserves direction and lets the model decide only how strongly to use it.

4 Relevance, Limitations, and Conclusion

The evaluated datasets emphasize built-environment change, so we do not claim biodiversity or ecosystem validation, although the prompt template and protocol are class-agnostic. RGEA is a screening aid, not a decision maker: the image score prioritizes pairs, the dense map directs inspection, r flags difficult acquisitions, a large P-AUC/P-F1 gap calls for recalibration, and a large I-AUC/P-AP gap for fixing geometry first. The model assumes approximate co-registration, uses one global reliability scalar, and ignores clouds, occlusion, and spatially varying registration; spatial reliability maps and non-building transitions come next. Signed evidence and its reliability must be modelled separately, and monitoring systems should report where the evidence lies, not only that change occurred.

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